

Huanqi Wen

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EDUCATION

- **Sun Yat-sen University** Guangzhou, China
M.S. in Mathematics 09/2023 - present
 - **GPA:** 89/100; 3.9/4;
 - **Advisor:** Prof. Lei Song
 - **Master's Thesis:** The N_p property of line bundles on toric varieties
- **Sun Yat-sen University** Guangzhou, China
B.S. in Mathematics, Minor in Economics 09/2019 - 06/2023
 - **Major GPA:** 89/100; 3.9/4;
 - **Undergraduate Thesis:** On the Minkowski sum of lattice points in lattice polytopes

RESEARCH INTERESTS

- **Pure mathematics:** toric geometry, syzygies of linear series; tropical geometry.
- **Applied mathematics:** formalization of mathematics, neuroscience and mathematics.

PUBLICATIONS AND PREPRINTS

- [1] Lei Song, Huanqi Wen, Zhixian Zhu. (2024). **On Covering Simplices by Dilations in Dimensions 3 and 4**, arXiv preprint, [arXiv:2404.02495](https://arxiv.org/abs/2404.02495)

RESEARCH EXPERIENCE

- **Positivity of line bundles on toric varieties** 09/2023 - present
With Prof. Lei Song
 - **Study** the case where the pair (X, L) of a toric variety X and a line bundle corresponding to a polytope P has the N_p properties.
 - **Theories/topics used:** stability of syzygy bundles, derived category of toric varieties, resolutions of toric subvarieties, Klyachko's classification of toric vector bundles.
 - **Computer tools:** SageMath; Macauley2.
- **Tropical geometry** 09/2023 - 01/2024
 - **Explored** the topic "Curve Counting Problems and Tropical Geometry"
 - **Wrote** a survey for a graduate-level course.
- **Minkowski sum of lattice points in lattice polytopes** 09/2022 - 09/2023
With Prof. Lei Song
 - **Studied** which lattice polytopes are normal, specifically whether the lattice points in an $n + 1$ dilation of a polytope P can be written as the sum of a lattice point in P and a lattice point in an n dilation of P .
 - **Original problem:** Oda's conjecture.
 - **Results:** by covering simplices with dilations, I proved that simplices with large lattice length in dimensions 3 and 4 are normal.
 - **Wrote** computer program (Python, matlab) to evaluate the normality of lattice polytopes with large lattice length.

READING SEMINARS

- **Speaker - Syzygies Seminar** 2025 Spring
Irena Peeva's book "Graded Syzygies"
- **Speaker - Toric Geometry Seminar** 2024 Spring - 2024 Fall
William Fulton's book "Introduction to Toric Varieties" and other classical papers [\[notes\]](#)
- **Algebraic Geometry Seminar** 2024 Spring - 2025 Spring
Robin Hartshorne's book "Algebraic Geometry"

CONFERENCES AND SUMMER SCHOOLS ATTENDED

- **2025 SRI in Algebraic Geometry** 07/2025
Colorado State University [[website](#)]
- **Stable Reduction of Algebraic Curves** 05/2025
Xiamen University, Speaker: Qing Liu
- **China Southeastern Algebraic Geometry Symposium VII** 12/2024
Sun Yat-sen University [[website](#)]
- **2024 International Summer school of Algebraic Geometry** 07/2024
Fudan University [[website](#)]

TEACHING

- **Sun Yat-sen University** 2025 Fall
Graduate teaching assistant of linear algebra.

AWARDS AND SCHOLARSHIPS

- **Graduate Scholarship (Top 20 %)** 2023-2024, 2024-2025, 2025-2026
Sun Yat-sen University
- **Honorable Mention, Mathematical Contest in Modeling (MCM)** 2022
Organized by the Consortium for Mathematics and Its Applications (COMAP)
- **Postgraduate Scholarship (Top 20 %)** 2020-2023, three times
Sun Yat-sen University

OTHER EXPERIENCE

- **International Scientific Research and Development Training** 2025 Spring
Completed the course on "Constructive Mathematics and Theoretical Computer Science," taught by Professor Vladimir Chernov.
- **Guangdong Digital Talent Training Program** 07/2022 - 08/2022
Conducted practical training and internships at China Telecom.
- **Department Student Union Member and Head of a Division** 09/2019 - 06/2021
Planned and led diverse student activities within the department

SKILLS

- Languages: English (IELTS: 7.0), Mandarin (native)
- Programming Languages: C++, Python
- Mathematical tools: MATLAB, SageMath, Macaulay2, $\text{\LaTeX}$